

IS311 Web Development

Week 07 - Lecture 08

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JavaScript

Object

- Represent real life objects
 - Properties
 - Methods

Object	Properties	Methods
	car.name = Fiat car.model = 500 car.weight = 850kg car.color = white	car.start() car.drive() car.brake() car.stop()

Object Properties

```
const car = {  
  name: "Mazda",  
  model: "Cx90",  
  color: "Platinum White"  
}
```

Property Name
Key

Property Value

Dot notation

```
car.name  
car.model  
car.color
```

Name Values

```
car["name"]  
car["model"]  
car["color"]
```

Accessing Object Properties

Object Methods

```
const car = {
  name: "Mazda",
  model: "Cx90",
  color: "Platinum White",
  move: function(){
    carImage = document.createElement("img");
    carImage.src = "images/car.png";
    carImage.style.width = "100px";
    carDiv = document.getElementById("carDiv");
    carDiv.appendChild(carImage);
  }
}
```

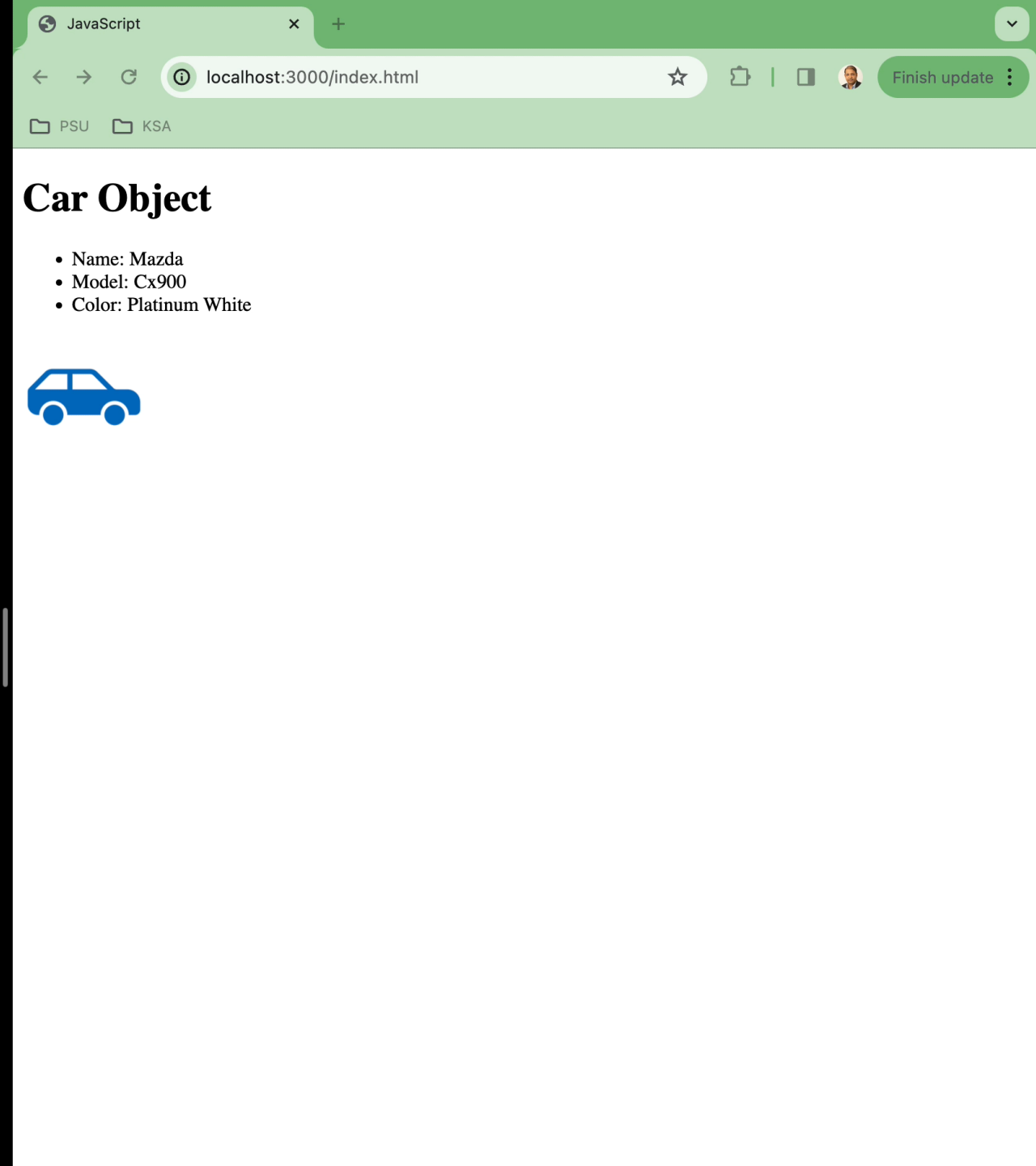
```
move = function(){
  carImage = document.createElement("img");
  carImage.src = "images/car.png";
  carImage.style.width = "100px";
  carDiv = document.getElementById("carDiv");
  carDiv.appendChild(carImage);
}
```

Accessing Object Method

- DOT notation `car.move();`
- Return function when used without parentheses

```
move_car = car.move;
```

```
index.html × # week07.css JS week07.js
index.html > html
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4     <meta charset="UTF-8">
5     <meta name="viewport" content="width=device-width">
6     <title>JavaScript</title>
7     <link rel="stylesheet" href="week07.css">
8     <script src="week07.js"></script>
9 </head>
10 <body>
11     <div id="carDiv" class="moveCar"></div>
12     <script>
13         car.move();
14     </script>
15 </body>
16 </html>
```



this Keyword

- In object method: Refers to **object**
- Alone refers to the **global object**
- In a function: refers to **global object**

this Keyword

- Refers to the object in object method

```
const person = {  
  firstName: "John",  
  lastName : "Doe",  
  id        : 5566,  
  fullName  : function() {  
    return this.firstName + " " + this.lastName;  
  }  
};
```


This Keyword

- Refers to the global object

[Learn More](#)

```

1  x = 78;
2
3  const car = {
4      name: "Mazda",
5      model: "Cx900",
6      color: "Platinum White",
7      move: function(){
8          carImage = document.createElement("img");
9          carImage.src = "images/car.png";
10         carImage.style.width = "100px";
11         carDiv = document.getElementById("carDiv");
12         carDiv.appendChild(carImage);
13     }
14 }
15
16 document.write("<h1>Car Object</h1>");
17 document.write("<ul>");
18 document.write("<li>Name: " + car.name + "</li>");
19 document.write("<li>Model: " + car["model"] + "</li>");
20 document.write("<li>Color: " + car.color + "</li>");
21 document.write("</ul>");
22
23 console.log("Value of x is ", this.x);

```

Nested Objects

```
person = {  
  firstName : "Abbas",  
  lastName : "Malik",  
  address : {  
    street: "Umer Bin Khattab Street",  
    district: "Sulaimania",  
    city: "Riyadh"  
  },  
  mobile: "0531864813"  
};
```

Object Constructor

▶ `{name: 'Facebook', country: 'USA'}`

• new Object()

```
company = new Object();
console.log(typeof company);
console.log(company);
```

object

▶ {}

```
company.name = 'Facebook';
company.country = 'USA';
console.log(company["name"]);
console.log(company["country"]);
```

Facebook

USA

Object Constructor

- `new Object()`

✖ ▶ Uncaught TypeError: Cannot set properties of undefined (setting 'firstName')
at [objects.js:24:23](#)

```
company.ceo.firstName = "Mark";
```

```
company.ceo = new Object();  
company.ceo.firstName = "Mark";
```

✖ Uncaught SyntaxError: Unexpected identifier 'name' (at [objects.js:27:18](#))

```
company.ceo.last name = "Zuckerberg";
```

```
company.ceo["last name"] = "Zuckerberg";
```

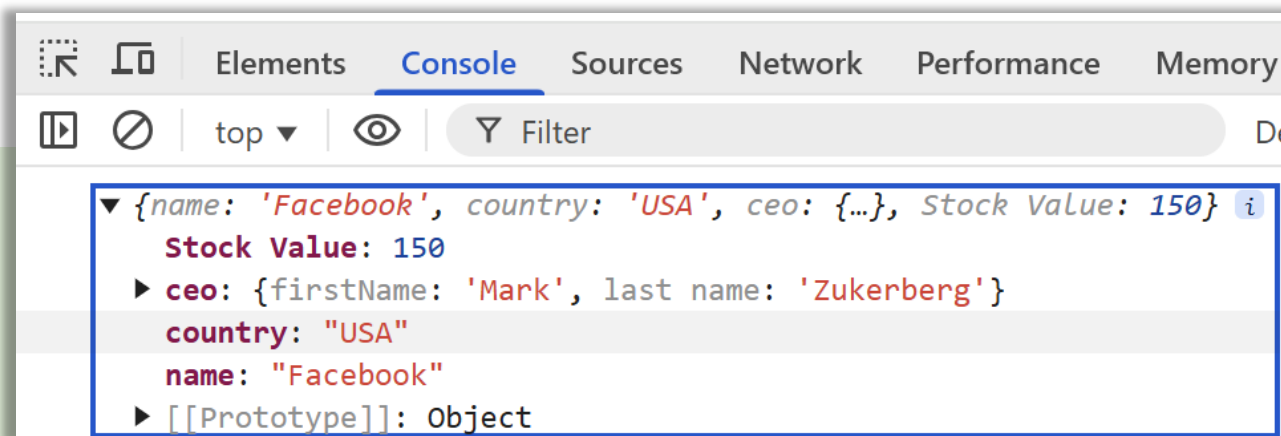
**Cannot use space
with DOT notation**

Object Constructor

```
company = new Object();
company.name = 'Facebook';
company.country = "USA";
```

```
company.ceo = new Object();
company.ceo.firstName = "Mark";
```

```
company.ceo["last name"] = "Zukerberg";
company["Stock Value"] = 150;
console.log(company);
```



String Concatenation

```
var pName = "Abbas";  
var greeting = "hello, " + pName + "!";  
console.log(greeting);
```

hello, Abbas!

Boolean

- All is false

NaN = Not a Number

```
if ( false && undefined && null && 0 && ' ' && "" && NaN)
    console.log("Value is true");
else
    console.log("All is false");
```

```
function multiply (x, y){
    return x * y;
}
console.log(multiply());
console.log(multiply(34));
console.log(multiply(34, 2));
```

NaN

NaN

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Boolean

- All is True

```
if ( true && 1 && -1 && "false" )  
  console.log("All is true")
```

Function and Default Values

```
function greet(name, greeting){
    console.log(name + ", " + greeting + "!");
}
```

```
greet();
greet("Abdulrahman");
greet("Abdulaziz", "Good evening");
```

```
undefined, undefined!
Abdulrahman, undefined!
Abdulaziz, Good evening!
```

```
greet("Abdulaziz", "Good evening", "additional!");
```

```
undefined, undefined!
Abdulrahman, undefined!
Abdulaziz, Good evening!
Abdulaziz, Good evening!
```

Function and Default Values

```
function greet(name, greeting){
  name = name || "Abbas";
  greeting = greeting || "Bonjour";
  console.log(name + ", " + greeting + "!");
}
```

```
greet();
greet("Abdulrahman");
greet("Abdulaziz", "Good evening");
greet("Abdulaziz", "Good evening", "additional!");
```

Abbas, Bonjour!

Abdulrahman, Bonjour!

Abdulaziz, Good evening!

Abdulaziz, Good evening!

Conditional Statements and Equality

```
var x = 4;  
var y = "4";  
  
if (x == y)  
    console.log("Value of x is 4 and y is '4'.");  
  
if (x === y)  
    console.log("Values of x and y as well as types are equal");  
else  
    console.log("Values of x and y as well as types are not equal");
```

for Loop

```
var sum = 0;
for(let i = 1; i <= 10; i++){
    console.log("Value of i is " + i);
    sum += i;
}
console.log("Sum of first 10 numbers is " + sum);
```

Value of i is 1
Value of i is 2
Value of i is 3
Value of i is 4
Value of i is 5
Value of i is 6
Value of i is 7
Value of i is 8
Value of i is 9
Value of i is 10
Sum of first 10 numbers is 55

for Loop

```
function multiplicationTable(num){
    num = num || 2;
    document.write("<div>");
    for(let i = 1; i <= 10; i++){
        document.write("<h2> " + num + " * " + i + " = " + num * i);
    }
    document.write("</div>");
}

multiplicationTable();
```

2 * 1 = 2
2 * 2 = 4
2 * 3 = 6
2 * 4 = 8
2 * 5 = 10
2 * 6 = 12
2 * 7 = 14
2 * 8 = 16
2 * 9 = 18
2 * 10 = 20

for in Loop

[More on loops in JS](#)

- Loop through properties of an Object

```
var person_names = {  
    name1 : "Abbas",  
    name2 : "Mohannad",  
    name3 : "Ali",  
    name4 : "Abdulrahman"  
};
```

```
for (let x in person_names){  
    document.write("<h2>" + person_names[x] + "</h2>");  
}
```

Abbas

Mohannad

Ali

Abdulrahman

Function is also an Object

```
var myFunction = function (){
    console.log("myFunction log");
}
```

```
myFunction.version = 1.01;
console.log(myFunction);
console.log(myFunction.version);
```

```
f (){
    console.log("myFunction log");
}
1.01
```


Array

- Holds more than one values

```
var myarray = new Array();
myarray[0] = 56;
myarray[1] = {
  name: "Abbas",
  role: "Lecturer",
  height: 5.10,
  age: 45
};
myarray[2] = 'string value';
myarray[3] = false;
console.log(myarray);
```

```
▼ (4) [56, {...}, 'string value', false] i
  0: 56
  1: {name: 'Abbas', role: 'Lecturer', height: 5.1, age: 45}
  2: "string value"
  3: false
  length: 4
  ▶ [[Prototype]]: Array(0)
```

Object

Array

```
array = [
  56,
  {
    name : "Abbas", role: "Lecturer",
    height: 5.10, age: 45
  },
  'string value',
  false
];
console.log(array);
```

▼ (4) [56, {...}, 'string value', false] **i**

- 0: 56
- ▶ 1: {name: 'Abbas', role: 'Lecturer', height: 5.1, age: 45}
- 2: "string value"
- 3: false
- length: 4
- ▶ [[Prototype]]: Array(0)

Object

Array Manipulation

```
cars = ['Toyota', 'Mazda', 'Hyundai', 'BMW', 'Geely'];  
for(let i = 0; i < cars.length; i++)  
    console.log(cars[i]);
```

Toyota

Mazda

Hyundai

BMW

Geely

Array is an Object

```
▼ (4) [56, {...}, 'string value', false] i
  0: 56
  ▶ 1: {name: 'Abbas', role: 'Lecturer', height: 5.1, age: 45}
  2: "string value"
  3: false
  length: 4
  ▶ [[Prototype]]: Array(0)
```

Object

```
array["value in array"] = "a value in array object";
```

```
▼ (4) [56, {...}, 'string value', false, value in array: 'a value in array object'] i
  0: 56
  ▶ 1: {name: 'Abbas', role: 'Lecturer', height: 5.1, age: 45}
  2: "string value"
  3: false
  value in array: "a value in array object"
  length: 4
  ▶ [[Prototype]]: Array(0)
```

Array Object with For Loop

```
array = [
  56,
  {
    name : "Abbas", role: "Lecturer",
    height: 5.10, age: 45
  },
  'string value',
  false
];
array["value in array"] = "a value in array object";
for(let i = 0; i < array.length; i++)
  console.log(array[i]);
```

```
▼ (4) [56, {...}, 'string value', false, value in array: 'a value in array object'] ⓘ
  0: 56
  1: {name: 'Abbas', role: 'Lecturer', height: 5.1, age: 45}
  2: "string value"
  3: false
  value in array: "a value in array object"
  length: 4
  [[Prototype]]: Array(0)
```

```
56
  ► {name: 'Abbas', role: 'Lecturer', height: 5.1, age: 45}
string value
false
```

Array Object with For Loop

```
array = [
  56,
  {
    name : "Abbas", role: "Lecturer",
    height: 5.10, age: 45
  },
  'string value',
  false
];
array["value in array"] = "a value in array object";
for (let i in array)
  console.log(array[i]);
```

```
▼ (4) [56, {...}, 'string value', false, value in array: 'a value in array object'] ⓘ
  0: 56
  ▶ 1: {name: 'Abbas', role: 'Lecturer', height: 5.1, age: 45}
  2: "string value"
  3: false
  value in array: "a value in array object"
  length: 4
  ▶ [[Prototype]]: Array(0)
```

```
56
▶ {name: 'Abbas', role: 'Lecturer', height: 5.1, age: 45}
string value
false
a value in array object
```

Array Properties and Methods

- Length Property: `<array_name>.length`
- `toString()`: converts an array to a string
- `join()`: join array elements into a string
- `pop()`: removes last element
- `push()`: add an element at the end
- `shift()`: remove first and shift others to lower index

[More on Methods](#)

Array Iteration

forEach()

```
const numbers = [45, 4, 9, 16, 25];  
let txt = "";  
numbers.forEach(myFunction);  
  
function myFunction(value, index, array) {  
    txt += value + "<br>";  
}
```


Array Iteration

map()

```
const numbers1 = [45, 4, 9, 16, 25];  
const numbers2 = numbers1.map(myFunction);  
  
function myFunction(value, index, array) {  
  return value * 2;  
}
```

Numbers2 = [90, 8, 18, 32, 50]

[More on array iteration](#)

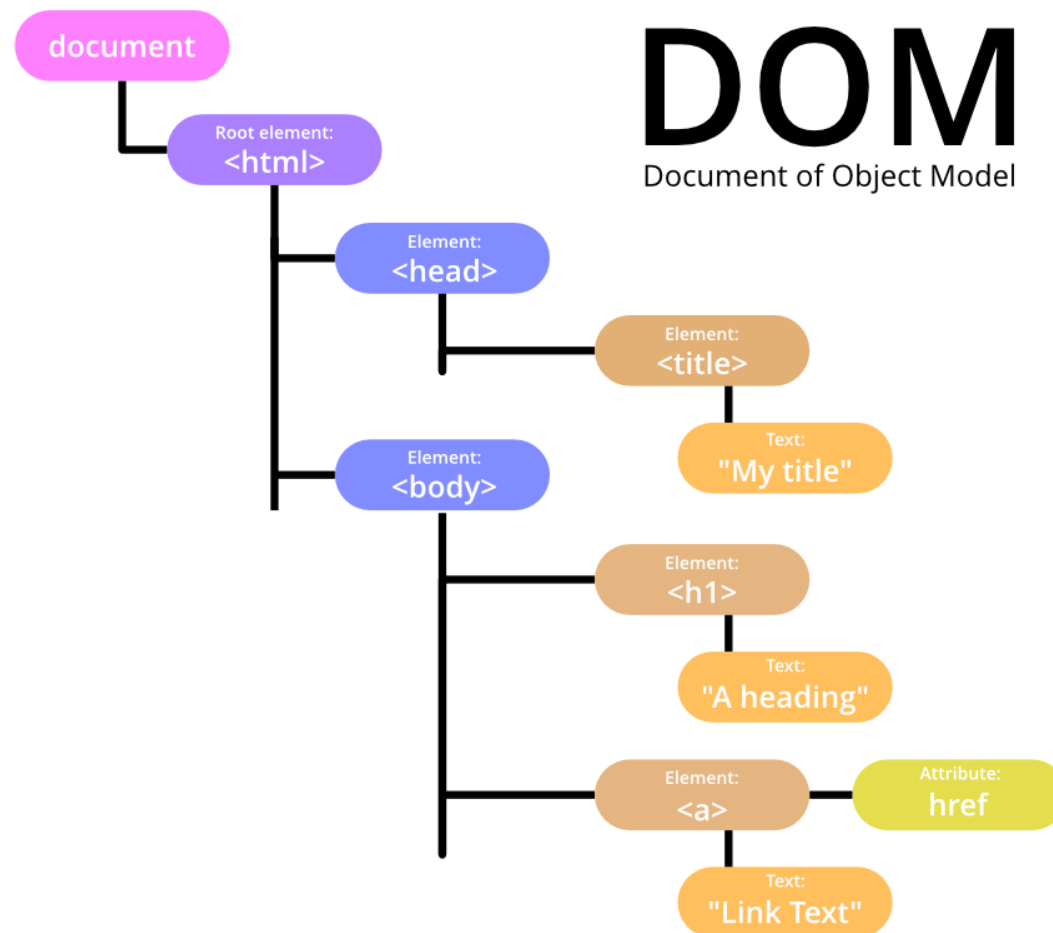
Array Operations

- Search an array
- Sort and array

HTML Document Object Model (DOM)

- A standard object model
- Programming interface for HTML documents
- A structured hierarchical representation of HTML document
- Easily access and manipulate tags, IDs, classes, attributes, or elements of HTML

HTML Document Object Model (DOM)

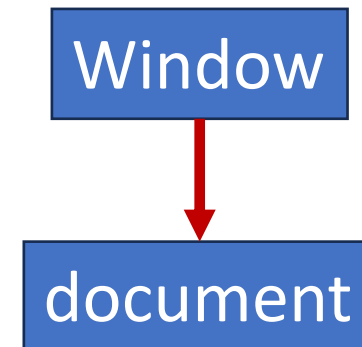


[Source](#)

DOM - Window = Global Context

- **this** represents global context **Window**

```
dom.html JS dom.js X
JS dom.js
1 console.log(this);
```



```
Window {window: Window, self: Window, document: document, name: 'Abbas<BS_START>{"bs":{"hardReload":true,"scroll":{"x":0,"y":0}}}<BS_START>', location: Location, ...} dom.js:1
```

DOM

```
console.log(document);
```

```
<> dom.html x JS dom.js
<> dom.html > ...
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport"
6          content="width=device-width, initial-scale=1.0">
7      <title>Document Object Model</title>
8      <script src="dom.js"></script>
9  </head>
10 <body>
11     <h1 id="header">Header of Document</h1>
12     <ul id="ulldist">
13         <li id="ulitem1">First Item</li>
14         <li id="ulitem2">Second Item</li>
15     </ul>
16     <div id="div">
17         <p id="paragraph">random text
18             <a id="linkpsu" href="http://www.psu.edu.sa">
19                 PSU
20             </a>
21         </p>
22     </div>
23 </body>
24 </html>
```

```
top Filter Default levels
▼ #document (http://localhost:3000/dom.html)
  <!DOCTYPE html>
  <html lang="en">
    ▼ <head>
      <meta charset="UTF-8">
      <meta name="viewport" content="width=device-width, initial-scale=1.0">
      <title>Document Object Model</title>
      <script src="dom.js"></script>
    </head>
    ▼ <body>
      ▶ <script id="__bs_script__"></script>
      <script async src="/browser-sync/browser-sync-client.js?v=3.0.2"></script>
      <h1 id="header">Header of Document</h1>
      ▼ <ul id="ulldist">
        ▶ <li id="ulitem1"></li>
        ▶ <li id="ulitem2"></li>
      </ul>
      ▼ <div id="div">
        ▼ <p id="paragraph">
          "random text "
          <a id="linkpsu" href="http://www.psu.edu.sa"> PSU </a>
        </p>
      </div>
    </body>
  </html>
```

DOM methods

`getElementById(elementId)`

```
console.log(document);  
var headertext = document.getElementById("header").textContent;
```

✖ ▶ Uncaught TypeError: Cannot read properties of null (reading 'textContent')
at dom.js:3:9

[DOM Properties and Methods](#)

DOM Events

- Page load
- Button clicked
- Hover over
- Blur
- Dblclick
- Beforeupload
- beforeprint
- ...

```
this.addEventListener("load", function (event) {  
    var headertext = document.getElementById("header")  
        .textContent;  
}  
);
```

[Complete List](#)

Handle onclick Event

- Let us add an input field and a button in page

```
<div>
  <label>Enter a Number: </label>
  <input id="tableNumber" type="number">
  <Button id="printTablebtn">Print Table</Button>
</div>
<div id="tableContent"></div>
```

onclick

```

dom.html JS dom.js x
JS dom.js > ...
1 console.log(document);
2 this.addEventListener("load", function (event) {
3     var headertext = document.getElementById("header")
4     .textContent;
5     var tablePrintBtn = this.document
6     .getElementById("printTablebtn");
7     tablePrintBtn.onclick = printTable;
8 }
9 );
10 function printTable(event){
11     let num = document.getElementById("tableNumber").value;
12     let tableHtml = "";
13     for(let i = 1; i <= 10; i++){
14         tableHtml += "<h2>" + num + " * " + i + " = "
15         + (i*num);
16     }
17     document.getElementById("tableContent")
18     .innerHTML = tableHtml;
19 }

```

```
dom.html JS dom.js x
JS dom.js > ...
1 console.log(document);
2 this.addEventListener("load", function (event) {
3     var headertext = document.getElementById("header")
4     .textContent;
5     var tablePrintBtn = this.document
6     .getElementById("printTablebtn");
7     tablePrintBtn.onclick = printTable;
8 }
9 );
10 function printTable(event){
11     let num = document.getElementById("tableNumber").value;
12     let tableHtml = "";
13     for(let i = 1; i <= 10; i++){
14         tableHtml += "<h2>" + num + " * " + i + " = "
15         + (i*num);
16     }
17     document.getElementById("tableContent")
18     .innerHTML = tableHtml;
19 }
20
```

Header of Document

- First Item
- Second Item

random text [PSU](#)

Enter a Number:

Elements Console Sources Network Performance Memory >>

top Filter Default levels No Issues 1

#document (http://localhost:3000/dom.html?) dom.js:1

Receiving Form Data - GET Method

- formpage.html file

Form Page

In this page using the following page, we will send the information to another HTML Page.

First Form: In this form, we will use the **GET** method to send data to *response.html* page.

First Name

Last Name

Date of Birth

Receiving Form Data - GET Method

- formpage.html file

```
<form action="responsepage.html" method="GET">  
  <label for="fName">First Name</label>  
  <input type="text" id="fName" name="first-name" /><br /><br />  
  <label for="lName">Last Name</label>  
  <input type="text" id="lName" name="last-name" /><br /><br />  
  <label for="dob">Date of Birth</label>  
  <input type="date" id="dob" name="dob" /><br /><br />  
  <input type="submit" value="submit" />  
</form>
```

Receiving Form Data - GET Method

- responsepage.html file



`http://127.0.0.1:5500/responsepage.html?first-name=Naif&last-name=Al-Majid&dob=2005-02-14`

Receiving Form Data

`http://127.0.0.1:5500/responsepage.html?first-name=Naif&last-name=Al-Majid&dob=2005-02-14`

```
<body>
  <div class="container">
    <h1>Response Page</h1>
    <table id="response-table" class="table">
      <thead>
        <tr>
          <th>First Name</th>
          <th>Last Name</th>
          <th>Date of Birth</th>
        </tr>
      </thead>
      <tbody id="table-body"></tbody>
    </table>
    <br />
    <a href="formpage.html">Go Back</a>
  </div>
  <script>
    const queryString = window.location.search;
    const urlParams = new URLSearchParams(queryString);
    html_string = `<tr>
      <td>${urlParams.get("first-name")}</td>
      <td>${urlParams.get("last-name")}</td>
      <td>${urlParams.get("dob")}</td>
    </tr>`;
    document.getElementById("table-body").innerHTML = html_string;
  </script>
```

Reference

- <https://www.w3schools.com/js/default.asp>
- <https://www.geeksforgeeks.org/dom-document-object-model/>
- https://developer.mozilla.org/en-US/docs/Web/API/Document_Object_Model
- <https://dom.spec.whatwg.org/>
- https://www.w3schools.com/jsref/dom_obj_event.asp